MIL-M-28787/432
AMENDMENT 1
29 September 1993

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MEMORY ARRAY, 128K DYNAMIC RANDOM ACCESS MODULE
KEY CODE JEJ

This amendment forms a part of Military Specification MIL-M-28787/432, dated 17 May 1991, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 4

TABLE I, Test conditions A1 and A2-A24: Add End-of-life limits below each line with a minimum limit of 1 MOhm (all temperatures) and a maximum limit of INF MOhm (all temperatures).

PAGE 6

TABLE I, Test conditions P1, P2, Test condition column: Delete "P1,P2" and substitute "P1-P8".

PAGES 80 AND 81

TABLE XV: Delete and substitute new table XV as printed on pages 3 through 6 of this amendment.

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TABLE XVI, Test conditions Q1-Q4, Maximum pattern rate: Delete "225" (four places) and substitute "320" (four places).

AMSC N/A
FSC 5963
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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (320 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (320 nS)	1-3
P3	Functional test Address complement test	1-5
P4	Functional test Marching columns test	1-4, 6
P5	Functional test March data test	1-4, 7
P6	Functional test Static refresh test	1-4, 8
P7	Functional test Refresh (cell retention) test	1-4, 9
P8	Functional test RAS only refresh test	1-4, 10

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test cond
Power supply, (Vcc)	4.5V 5.5V 5.0V	± 45 mV ± 55 mV ± 50 mV	P1 P2 P3-P8
Power supply current limit	1.41 A	NA	P1-P8
Pattern voltage high, (VH) low, (VL) high, (VH) low, (VL)	 2.4 V 0.4 V 3.0 V 0.0 V	 ± 100 mV ± 100 mV ± 100 mV ± 100 mV	 P1,P2 P1,P2 P3-P8 P3-P8

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NOTES: - Continued.

2. The following procedure is required to perform this test:
(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: 2.4 V ≤ Voh ≤ Vcc
Low level: 0.0 V ≤ Vol ≤ 0.5 V

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

4. Memory must be initialized before testing. Initialization is performed by running through 8 RAS cycles (256 rows each cycle) to charge the memory cells. If any pattern is performed immediately after another, there is no need to initialize memory before each pattern. If refresh must be halted for any time greater than 4 mS between patterns, the memory must be initialized.

5. Address complement test.
(a) Load memory with all "0's".
(b) Read minimum address location.
(c) Load minimum address location with "1".
(d) Read maximum address location.
(e) Load maximum address location with "1".
(f) Read minimum address location + 1.
(g) Load minimum address location + 1 with "1".
(h) Read maximum address location - 1.
(i) Load maximum address location - 1 with "1".
(j) Repeat (g) through (i) until all memory locations are loaded with "1's".
(k) Repeat (b) through (j) reading "1" and loading "0" in all memory locations.

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NOTES: - Continued.

6. Marching columns test.
   (a) Load all memory locations with "0".
   (b) Write "1" into all locations in column 1.
   (c) Read "1" in all locations in column 1.
   (d) Write "0" in all locations in column 1.
   (e) Repeat (b) through (d) for all columns 2-n sequentially.
   (f) Load all memory locations with "1".
   (g) Write "0" in all locations in column 1.
   (h) Read "0" in all locations in column 1.
   (i) Write "1" in all locations in column 1.
   (j) Repeat (g) through (i) for columns 2-n sequentially.

7. March data test.
   (a) Load memory with all "0's".
   (b) Read location 0.
   (c) Write "1" into location 0.
   (d) Read "1" in location 0.
   (e) Repeat (b) through (d) for all other locations sequentially.
   (f) Read "1" in maximum address location.
   (g) Write "0" in maximum address location.
   (h) Read "0" in maximum address location.
   (i) Repeat (f) through (h) for all other locations sequentially.
   (j) Load memory with all "1's".
   (k) Read minimum address location.
   (l) Write minimum address location with "0".
   (m) Read "0" in minimum address location.
   (n) Repeat (k) through (m) for all other locations sequentially.
   (o) Read "0" in maximum address location.
   (p) Load "1's" into maximum address location.
   (q) Read "1's" in maximum address location.
   (r) Repeat (o) through (q) for all other locations sequentially.

8. Static refresh test.
   *This test performed only at 85°C.
   (a) Load memory with all "0's".
   (b) Pause 2 ms.
   (c) Read memory with all "0's".
   (d) Load memory with all "1's".
   (e) Pause 2 ms.
   (f) Read memory with all "1's".

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NOTES: - Continued.

9. Refresh test (cell retention).
   (a) Load memory with all "0's".
   (b) Pause 1 ms.
   (c) Read memory with all "0's".
   (d) Repeat (a) through (c) using all "1's".
   (e) Read minimum address location.
   (f) Read maximum address location.
   (g) Repeat (e) and (f) for 1 mS.
   (h) Read memory with all "1's".
   (i) Load memory with all "0's".
   (j) Repeat (e) through (h) with all "0's".
   (k) Read address location 32767.
   (l) Read address location 32768.
   (m) Repeat (k) and (l) for 1 ms.
   (n) Read memory with all "0's".
   (o) Load memory with all "1's".
   (p) Repeat (k) through (n) with all "1's".

10. RAS only refresh test at 85°C.
   (a) Load memory with all "0's".
   (b) Perform 128 RAS cycles (CAS = WE = "1").
   (c) Repeat (b) for 0.5 second.
   (d) Read memory with all "0's".
   (e) Repeat (a) through (d) with all "1's".

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